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(Helianthus),a few layers of sclerenchyma cell occur in patches outside the phloem in each vascular bundle. This patch of sclerenchyma cell is called **Bundle cap or Hardbast**. The bundle caps and the parenchyma cells between them constitute the pericycle in the stem of sunflower.

Vascular Bundles

The vascular bundles consist of xylem, phloem and cambium. Xylem and phloem in the stem occur together and form the vascular bundles. These vascular bundles are **Wedge shaped**. They are arranged in the form of a ring. Each vascular bundle is **conjoint, collateral, open and endarch**.

Phloem

Primary phloem lies towards the periphery. It consists of **protophloem and metaphloem.** Phloem consists of sieve tubes, companion cells and phloem parenchyma. Phloem fibres are absent in the primary phloem. Phloem conducts organic food materials from the leaves to other parts of the plant body.

Cambium

Cambium consists of **brick shaped** and thin walled meristematic cells. It is one to four layers in thickness. These cells are capable of forming new cells during **secondary growth**.

Xylem

Xylem consists of xylem fibres, xylem parrenchyma vessels and tracheids. Vessels are thick walled and arranged in a few rows.

Xylem conducts water and minerals from the root to the other parts of the plant body.

Pith

The large central portion of the stem is called **pith**. It is composed of parenchyma cells with intercellular spaces. The pith is also known as **medulla**. The pith extends between the vascular bundles. These extensions of the pith between the vascular bundles are called primary pith rays or primary medullary rays. Function of the pith is **storage of food.**

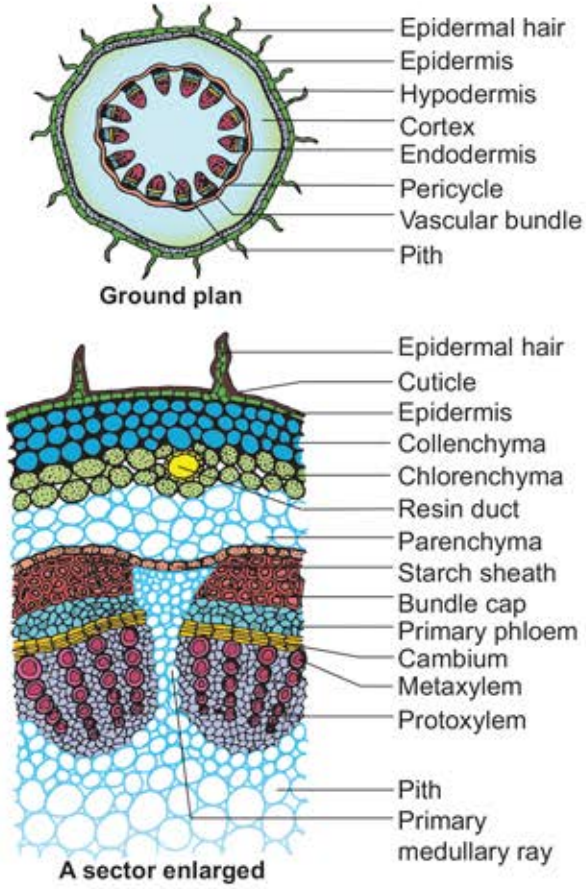


Figure 9.21: T. S of Dicot Stem (Sunflower stem)

Primary Structure of Monocot Stem-maize Stem

The outline of the maize in transverse section is more or less circular. The transverse section of the monocot stem [maize] shows the following plan of arrangement of tissues from the periphery to the centre.